

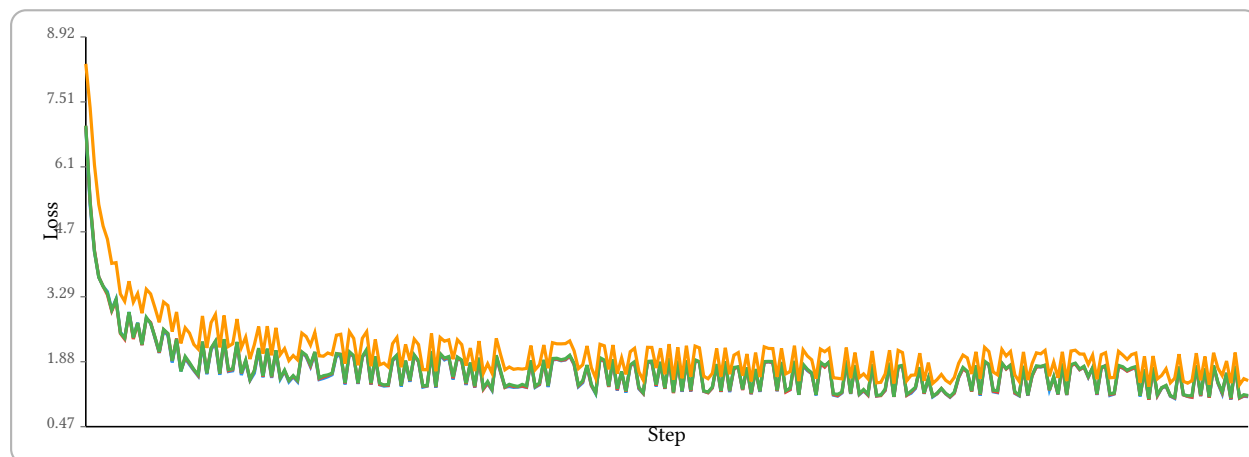
TinyStories Head Sweep: GIANT vs TiDAR

Runs stopped manually after about 1.5h. Each curve is plotted up to its last logged step.

- h6 MHA (baseline)
- h18 MHA
- h18 with kv6 GQA
- h6 MHA (head_dim=32 control)

GIANT training loss

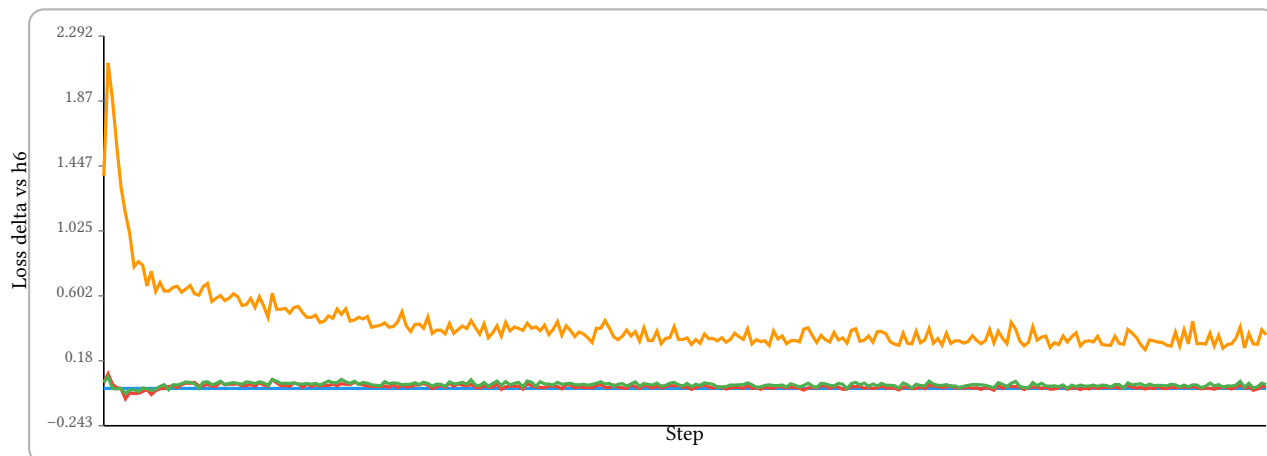
Interpretation: lower is better.



Logged points: h6 270, h18 270, gqa 270, h6hd32 270. End steps: h6 54000, h18 54000, gqa 54000, h6hd32 54000.

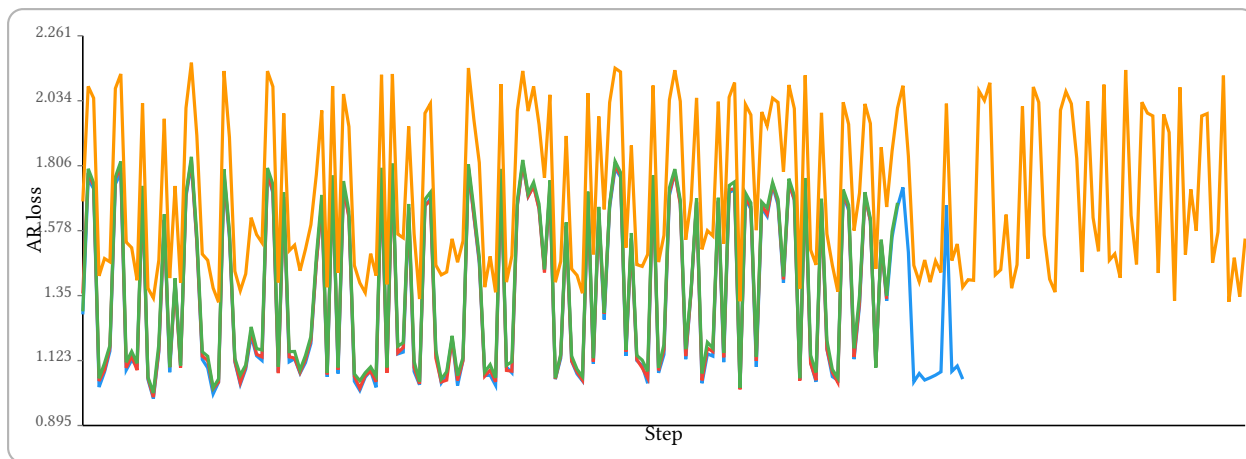
GIANT loss delta vs baseline

Interpretation: baseline is 0.00. Below 0.00 is better than baseline, above 0.00 is worse.



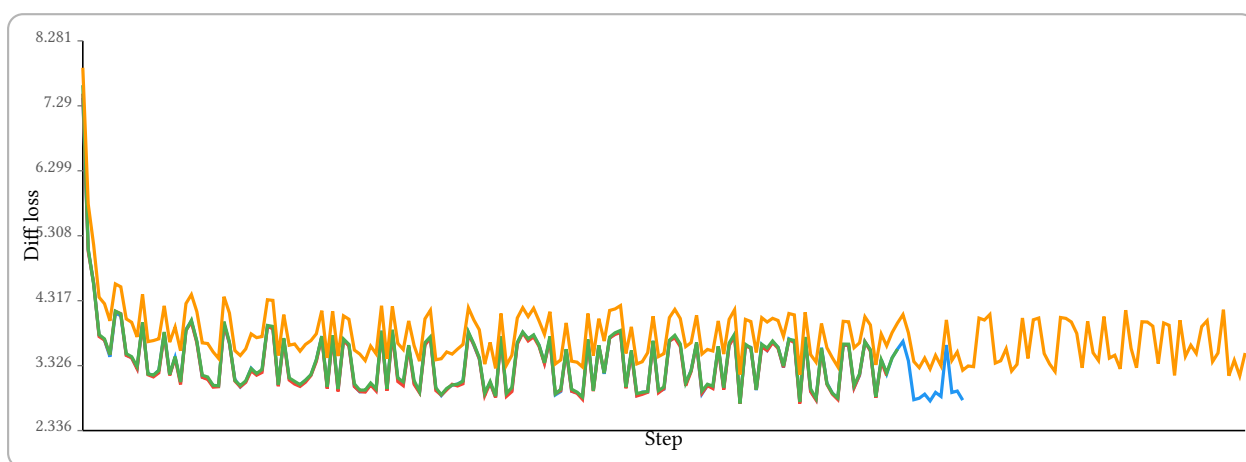
TiDAR AR loss

Interpretation: lower is better.



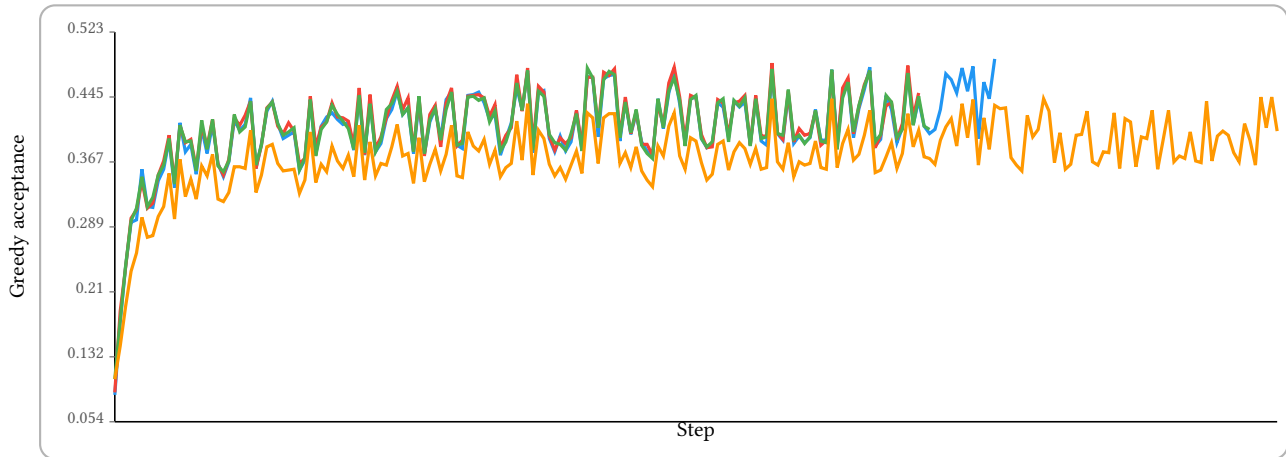
TiDAR Diff loss

Interpretation: lower is better.



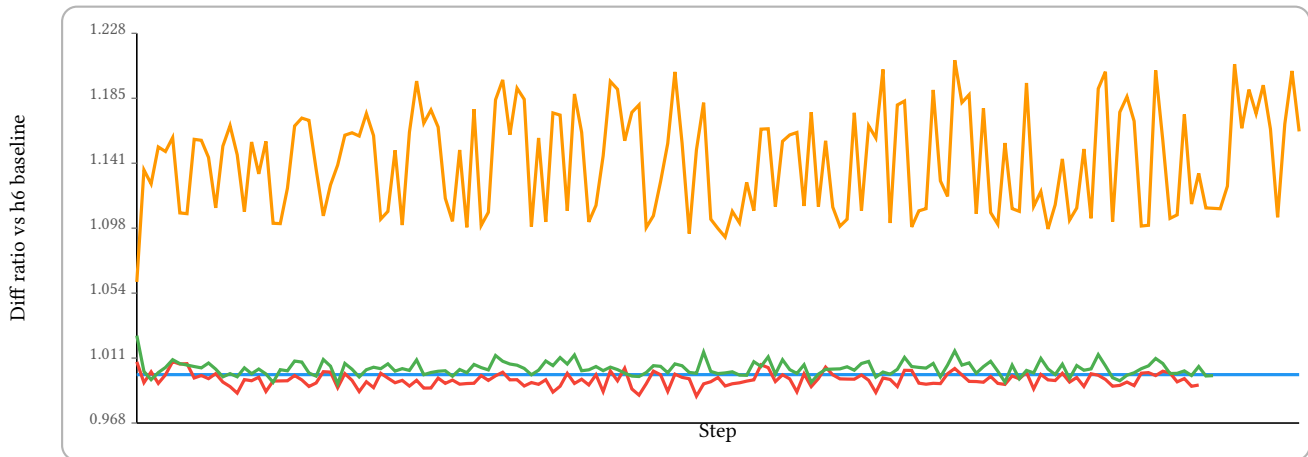
TiDAR Greedy acceptance

Greedy acceptance = fraction of token positions where $\text{argmax}(\text{AR})$ equals $\text{argmax}(\text{Diff})$. Interpretation: higher is better.



TiDAR Diff delta vs baseline (normalized)

Baseline is h6 MHA fixed at 1.00. Other lines are ratio = $\text{Diff}(\text{loss_run}) / \text{Diff}(\text{loss_baseline})$ at matching steps. Interpretation: 1.00 matches baseline; below 1.00 is better than baseline; above 1.00 is worse than baseline.



Ratio points: h18 149, gqa 151, h6hd32 163.

Appendix: Updated sweep interpretation

GIANT summary (full 54k-step epoch) Runtime vs h6 baseline (1797.2s): h18 MHA = +3.66% (1863.0s), h18 kv6 GQA = -3.10% (1741.4s), h6 hd32 = -63.0% (665.0s). Final GIANT loss at step 54k: h6 = 1.1174, h18 = 1.1297, h18 kv6 GQA = 1.1423, h6 hd32 = 1.4661.

TiDAR summary (timed runs) Reached steps: h6 = 32.4k, h18 = 29.6k, h18 kv6 GQA = 30.0k, h6 hd32 = 42.8k. Diff ratio vs h6 baseline (mean over matched steps): h18 = 0.9966, h18 kv6 GQA = 1.0039, h6 hd32 = 1.1425.

Short interpretation Within the comparable-width models (h6/h18/GQA at $d_{\text{model}}=576$), more heads gave little quality gain for the extra cost: h18 is slightly slower, and GQA is near-parity on quality with modest speed benefit. The h6 hd32 control buys much higher throughput, but quality drops clearly; it is a compute-for-quality trade, not a strict heads-only comparison.